MediScan AI - Business Proposal

Executive Summary

MediScan AI is an artificial intelligence platform designed to assist hospitals in diagnosing diseases from X-ray, CT scan, and MRI images. The system leverages deep learning models to improve diagnostic accuracy, reduce reporting time, and support healthcare professionals in making faster clinical decisions.

Problem Statement

Many hospitals experience delays in medical image analysis due to a shortage of radiologists and increasing patient volumes. Early detection of diseases is often delayed, impacting patient outcomes.

Proposed Solution

Develop a cloud-based AI platform capable of:

- Detecting abnormalities in medical images.
- Providing confidence scores for diagnoses.
- Generating preliminary diagnostic reports.
- Integrating with existing hospital information systems (HIS).

Target Market

- Medium and large hospitals
- Diagnostic imaging centers
- Telemedicine providers
- Government healthcare institutions

Financial Overview

Initial Investment: $500,000

Projected Revenue:

- Year 1: $300,000
- Year 2: $900,000
- Year 3: $2,000,000

Estimated Break-even: Approximately 30 months after launch.

Technology Stack

- Python
- PyTorch / TensorFlow
- FastAPI Backend

- React Frontend
- PostgreSQL Database
- AWS Cloud Infrastructure
- Docker & Kubernetes

Risk Assessment

Technical Risks:

- Model accuracy
- Data quality
- Large-scale deployment

Business Risks:

- Competition
- Customer acquisition costs
- Long sales cycle

Regulatory Risks:

- Healthcare data privacy
- HIPAA/GDPR compliance
- Medical device certification requirements

SWOT Analysis

Strengths:

- AI-assisted diagnostics
- Faster reporting
- Scalable cloud architecture

Weaknesses:

- High initial development cost
- Requires large annotated datasets

Opportunities:

- Growing AI adoption in healthcare
- Expansion into global markets
- Integration with telemedicine

Threats:

- Established healthcare AI competitors
- Regulatory changes
- Cybersecurity threats

Evaluation Request

The executive board is requested to evaluate: 1. Technical feasibility 2. Financial viability 3. Market opportunity 4. Regulatory compliance 5. Operational risks 6. Scalability 7. Overall investment recommendation